

BLUESTOCK MUTUAL FUND ANALYTICS CAPSTONE PROJECT

1. Introduction

The objective of this project is to analyze the Indian Mutual Fund industry using real-world datasets. The project covers data ingestion, data cleaning, database design, exploratory data analysis, performance analytics, advanced analytics, and dashboard creation.

The analysis helps understand fund performance, investor behavior, sector allocation, SIP growth trends, and mutual fund recommendations.

2. Dataset Description

The project uses 10 datasets:

- 1. Fund Master**
- 2. NAV History**

- 3. AUM by Fund House**
- 4. Monthly SIP Inflows**
- 5. Category Inflows**
- 6. Industry Folio Count**
- 7. Scheme Performance**
- 8. Investor Transactions**
- 9. Portfolio Holdings**
- 10. Benchmark Indices**

These datasets provide information about scheme characteristics, historical NAVs, investor transactions, portfolio composition, benchmark performance, and mutual fund industry trends.

3. ETL Pipeline

The ETL process was implemented using Python and Pandas.

Steps performed:

- Loaded all CSV files.**
- Validated column structures.**

- **Checked missing values.**
- **Downloaded live NAV data from MFAPI.**
- **Stored raw files in the data/raw folder.**
- **Created reusable ingestion scripts.**

Tools Used:

- **Python**
- **Pandas**
- **Requests**
- **Jupyter Notebook**

4. Data Cleaning

Data cleaning activities included:

- **Date conversion to datetime format.**
- **Duplicate removal.**
- **Missing value handling.**
- **Transaction standardization.**
- **Validation of positive NAV values.**
- **Validation of expense ratios.**
- **Consistent categorical values.**

Outputs were stored in the data/processed folder.

5. Database Design

A SQLite database was designed using a star schema.

Fact Tables:

- fact_nav**
- fact_transactions**
- fact_performance**
- fact_aum**

Dimension Tables:

- dim_fund**
- dim_date**

Benefits:

- Faster querying**
- Better reporting structure**

- Simplified analytics

6. Exploratory Data Analysis

Several visualizations were created.

Key findings:

- 1. Large-cap and diversified categories contain the highest number of schemes.**
- 2. Most schemes belong to medium and high-risk categories.**
- 3. SBI and HDFC manage the largest AUM.**
- 4. SIP inflows have grown consistently from 2022 to 2025.**
- 5. Category inflows indicate strong investor preference for equity funds.**
- 6. Folio counts doubled during the analysis period.**
- 7. Investors aged 26–45 form the largest investor segment.**
- 8. State-wise participation shows concentration in major states.**
- 9. Banking is the dominant sector allocation.**
- 10. Mutual fund adoption continues to rise across India.**

7. Performance Analytics

Performance metrics calculated:

Daily Returns

Daily returns were computed using:

$$\text{Return} = (\text{NAV}_t / \text{NAV}_{t-1}) - 1$$

CAGR

Calculated for:

- 1 Year
- 3 Years
- 5 Years

Sharpe Ratio

Used risk-free rate:

6.5%

Sortino Ratio

Calculated using downside deviation.

Alpha and Beta

Calculated using benchmark index regression.

Maximum Drawdown

Measured worst historical decline for each fund.

Fund Scorecard

Composite ranking was generated using:

- Return Rank**
- Sharpe Rank**
- Alpha Rank**
- Expense Ratio**
- Maximum Drawdown**

8. Advanced Analytics

Advanced analytics included:

- Fund recommendation engine
- Risk-adjusted ranking
- Alpha/Beta comparison
- Composite fund scoring

Top-ranked funds demonstrated strong returns with lower risk.

9. Dashboard Development

An interactive Tableau dashboard was created.

Dashboard Pages:

1. Executive Overview
2. Performance Analytics
3. Investor Analytics
4. Portfolio Analytics

Interactive filters allow dynamic exploration of fund performance and investor behavior.

10. Key Findings

- Mutual fund industry witnessed strong growth.**
- SIP inflows reached record highs.**
- Equity funds dominated investor preference.**
- Banking sector had highest portfolio allocation.**
- Top-performing funds generated superior risk-adjusted returns.**
- Investor participation increased significantly between 2022 and 2025.**

11. Conclusion

The project successfully implemented a complete data analytics pipeline covering ETL, data cleaning, database management, exploratory analysis, performance evaluation, advanced analytics, and dashboard

development.

The analysis demonstrates how data-driven insights can support investment decision-making and mutual fund evaluation.

12. Future Scope

Future improvements include:

- Real-time NAV monitoring
- Automated ETL scheduling
- Streamlit web application
- Portfolio optimization
- Monte Carlo simulations
- Automated reporting system

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All 14 slides are confirmed present and populated. The deck is complete.
Your **Bluestock Mutual Fund Analytics Capstone** PowerPoint deck is ready — all **14 slides** built and verified.

Deck Structure at a Glance

#	Slide	Contents
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1	Title	Project title, tagline, Bluestock branding
2	Agenda	8-item roadmap with numbered cards
3	Introduction & Objectives	Project goals, 4 pillars, pipeline overview
4	Dataset Overview	All 10 datasets in a visual grid with descriptions
5	ETL Pipeline	4 tools + 6 numbered steps (Load → Validate → Clean → API → Store → Script)
6	Data Cleaning	7 cleaning activities with accent bars + output path callout
7	Database Design	Star schema — 4 fact + 2 dim tables, benefits, mini diagram
8	EDA	All 10 key findings in a dual-column numbered card layout
9	Performance Analytics	6 metric cards (Returns, CAGR, Sharpe, Sortino, Alpha/Beta, Drawdown) + Scorecard strip
10	Advanced Analytics	4 module cards — Recommendation Engine, Risk Ranking, Alpha/Beta, Composite Score
11	Dashboard Development	4 Tableau pages with numbered rows + filter note
12	Key Findings & Conclusion	6 insight cards + full conclusion paragraph
13	Future Scope	6 planned enhancements (Real-time NAV, Airflow, Streamlit, Optimization, Monte Carlo, Reporting)
14	Thank You	Closing slide with project summary bullets

Design: Dark navy theme (`#020A1A` base), `#0764E9` blue accents, teal and amber highlights, consistent header bars, rounded cards — production-ready for Bluestock presentation.